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Muscle-fiber pennation angles are greater in hypertrophied than in normal muscles

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KAWAKAMI, YASUO, TAKASHI ABE, AND TETSUO FUKUNAGA. *Muscle-fiber pennation angles are greater in hypertrophied than in normal muscles.* J. Appl. Physiol. 74(6): 2740–2744, 1993.—Muscle-fiber pennation angles were measured in vivo with the use of ultrasonography to investigate the relationship between fiber pennation and muscle size for 32 male subjects (from untrained subjects to highly trained bodybuilders). From the image of a B-mode ultrasonogram, fiber pennation angles and thickness of triceps brachii were determined, the former as angles between echoes from the interspaces of fascicles and from the aponeurosis of long and medial heads of triceps and the latter as the distance between the fat-muscle and muscle-bone interfaces. The pennation angles were in the range of 15 and 53° for the long head and 9 to 26° for the medial head, which were similar to or greater than the published and the present data on human cadavers. Significant differences were observed between normal subjects and bodybuilders in muscle thickness and pennation angles ($P < 0.01$), and there were significant correlations between muscle thickness and pennation angles for both long ($r = 0.884$) and medial ($r = 0.833$) heads of triceps, suggesting that muscle hypertrophy involves an increase in fiber pennation angles.

ultrasound; muscle thickness; pennate muscle; triceps brachii

MORPHOLOGICALLY, skeletal muscles are roughly divided into two types, parallel fiber muscles in which fibers lie almost parallel to the line of action of the muscle and pennate muscles in which fibers insert into tendons at an angle to the line of action of the muscle (10). Pennate muscles have received much interest in terms of their functional characteristics (3, 7, 17).

The increase in muscular strength through resistance training has been attributed to muscle hypertrophy, neural adaptation, and changes in muscle fiber architecture (4, 9, 10, 14). Narici et al. (14) concluded that muscle hypertrophy explained 40% of the increase in muscle strength, and the other 60% was attributed to changes in neural drive, muscle architecture, or both. Maughan et al. (12) demonstrated an inverse relationship between muscle cross-sectional area (CSA) and the ratio of muscle strength to muscle CSA, and they suggested that in larger muscles pennation angles might be greater, resulting in a smaller strength-to-CSA ratio.

Recent progress in technology has made possible muscle CSA measurement in vivo with the use of ultrasonography (8), computed tomography (12), and magnetic resonance imaging (14), whereas, to date, information on muscle architecture (pennation angles, fiber length) has

been derived mainly from the study on human cadavers (1, 2, 16). Recently, Henriksson-Larsen et al. (6) proposed a method of measuring pennation angles and muscle-fiber length in vivo with the use of an ultrasound apparatus. They showed that fiber angulation of the vastus lateralis muscle was independent of fiber size, which was contradictory to the previous studies cited above.

The purpose of the present study was to measure fiber pennation angles in vivo with the use of ultrasonography to investigate the relationship between pennation angles and muscle thickness in subjects of different training statuses, with the hypothesis that larger muscles have greater pennation angles.

METHODS

Subjects. Thirty-two healthy males [age, 23 ± 5 yr (SD); ht, 171 ± 6 cm; mass, 70 ± 11 kg] volunteered for the study. Informed consent was obtained from each subject before the study began. The subjects differed in their training status and included untrained undergraduate students, moderately active subjects, and highly trained bodybuilders. Upper arm circumference of the subjects ranged from 24.8 to 40.5 cm.

Measurement of muscle thickness. Muscle thickness of triceps brachii was measured with a B-mode ultrasound apparatus (SSD-500, Aloka). Precision and linearity of the image were confirmed beforehand by using an acoustic standoff (model AC-2, ATS Laboratories) with pins stuck diagonally in the image and by comparing the distance between pins with the vertical distance between the echoes of the pins in the reconstructed image. The error was calculated by

$$\text{error} = (\text{measured distance} - \text{actual distance}) / \text{actual distance}$$

The error observed at the superficial and deep areas of the image was the same (1.3%), and the distance measurement from the reconstructed image was considered reliable.

Our method of measuring muscle thickness was similar to the procedures of Weiss and Clark (15) and Martinson and Stokes (11). A single cross-sectional plane was measured at a site 40% of the distance from the lateral epicondyle to the acromion process of the scapula, starting at the lateral epicondyle. The muscles measured included parts of the long and medial heads of the triceps brachii. The subjects stood with their arms relaxed in

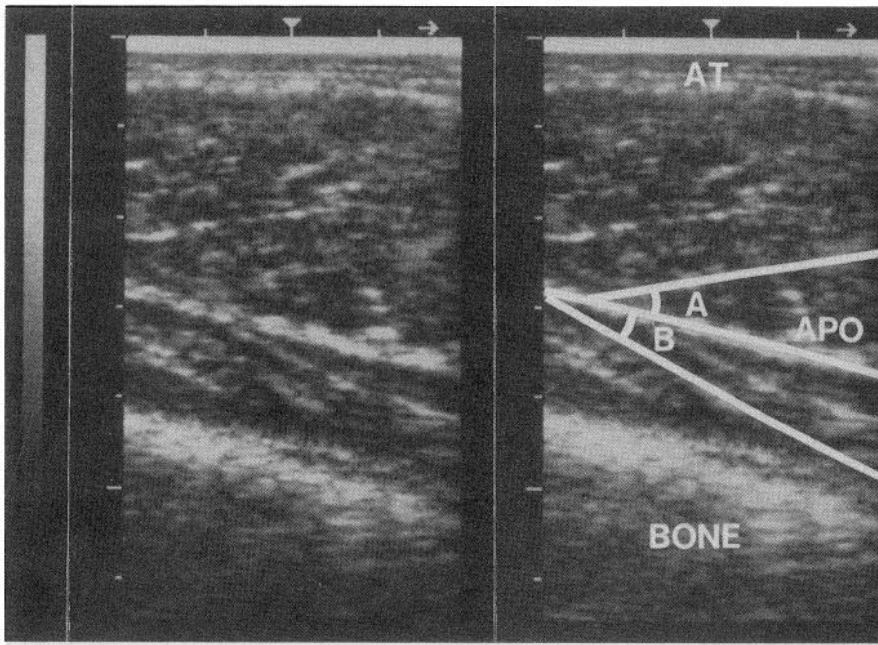


FIG. 1. Ultrasonographic image from 1 subject (left) and same image in which white lines indicate echoes from aponeurosis (APO) and interspaces among fascicles (right). Angles *A* and *B* were defined as fiber pennation angles of long and medial heads, respectively. AT, subcutaneous adipose tissue, Bone, humerus.

extended position. A transducer with a 5-MHz scanning head was placed perpendicular to the tissue interface and perpendicular to the underlying humerus. The scanning head was coated with water-soluble transmission gel, which provided acoustic contact without depressing the dermal surface. The subcutaneous adipose tissue-muscle interface and the muscle-bone interface were identified from the ultrasonic image, and the distance from the adipose tissue-muscle interface to the muscle-bone interface was adopted as representative of muscle thickness. Both the long and medial heads of the triceps were included in the muscle layer.

Measurement of fiber pennation angles. The center of the transducer of the ultrasonic apparatus was placed at the same site where the muscle thickness measurements were taken, this time parallel to the long head of the triceps. Before the measurement, the subject extended his elbow and exerted slight isometric force. The tester visually confirmed the muscle belly of the long head and drew a longitudinal line on the dermal surface in the center of the belly with a surgical marking pen. The angles between the echoes of the aponeurosis of the triceps and the echoes from interspaces among the fascicles were measured and defined as muscle-fiber pennation angles (Fig. 1). The echoes from interspaces of the medial head of the triceps were sometimes imaged more clearly when the plane was changed slightly diagonally to the line drawn with the pen, in which case the recreated image was used for the measurement of pennation angles of the medial head. Eleven of the 32 subjects were randomly selected and tested twice for reproducibility of measurement.

To validate the determination of muscle thickness and pennation angles, the ultrasound measurement was carried out on the limbs of three human cadavers [two males (73 and 84 yr old) and one female (75 yr old)]. After dissection, the long head of the triceps was divided along the long axis of the muscle, and muscle thickness and pennation angles were measured manually, the former as

the distance between the superficial layer of the long head and the humerus and the latter as the angles between the aponeurosis and the fascicles. The ultrasonic transducer was placed on the surface of the long head, and in the reconstructed image the angles between the echoes of the fascicles and the aponeurosis and the muscle thickness of both the long and medial heads were measured and compared with the manually determined pennation angles and muscle thickness. Pennation angles were measured at two positions, and the averaged value of the two angles was used in both the manual and the ultrasonic measurements. Those measurements were carried out separately by two persons testing blindly.

Statistical analyses. Reproducibility of measurement of muscle thickness and muscle-fiber pennation angles was tested by regression analyses and Student's *t* test. Statistical significance of the relationship between muscle thickness and physical measures (body mass and upper arm circumference) was studied by regression analysis, and Pearson product-moment correlations (*r*) were calculated. Differences in muscle thickness and fiber pennation angles between normal subjects and bodybuilders were tested by unpaired *t* test. Statistical significance of the relationship between muscle thickness and pennation angles was checked by the regression analyses. Level of significance was accepted at $P < 0.05$.

RESULTS

Table 1 shows manual and ultrasonic measurements of muscle thickness and pennation angles on human cadavers. Muscle thickness and pennation angles ranged from 12 to 21 mm and from 9 to 16°, respectively. Ultrasonic measurements differed from manual measurements by 0–1 mm for muscle thickness and by 0–1° for pennation angles.

In *in vivo* measurements on 32 subjects, muscle thickness ranged from 28 to 61 mm. The correlation coefficient between the first and second measurements of

TABLE 1. Manual and ultrasonic measurements of muscle thickness and fiber pennation angles of triceps brachii

	Age, yr	Sex	Limb Side	Muscle Thickness, mm		Pennation Angle, degrees	
				Manual	Ultrasonic	Manual	Ultrasonic
Cadaver 1	73	M	Right	19	19	14	14
Cadaver 2	84	M	Left	21	20	16	15
			Right	18	19	10	11
Cadaver 3	75	F	Left	13	12	9	10

Muscle thicknesses are for both long and medial heads. Pennation angles are for long head. M, male; F, female.

muscle thickness for 11 subjects was $r = 0.978$. There was no significant difference in muscle thickness between the mean values of the two measurements. Statistically significant relationships were observed between muscle thickness and body mass ($r = 0.790$) and between muscle thickness and upper arm circumference ($r = 0.882$).

As shown in Fig. 1, echoes from the interspaces among the fascicles could be recognized for all subjects. The angles between the aponeurosis and the fascicles (fiber pennation angles) differed slightly (0.5 – 1°) in one ultrasonic image; thus, the averaged value of three measurements from different parts of a single image was considered representative of the fiber pennation angle. The average values were in the range of 15 and 53° for the long head and 9 and 26° for the medial head, which were similar to or greater than those of human cadavers (long head 9° ; medial head, 13°) (1) and the present cadaver data (long head, 9 – 16°). Two measurements from 11 subjects gave correlation coefficients of $r = 0.947$ for the long head and $r = 0.865$ for the medial head, with no significant difference between the two mean values.

The lines between the fascicles were linear in most subjects, whereas in well-trained bodybuilders with upper arm circumference ~ 40 cm the lines were curvilinear, arising from the aponeurosis at steeper angles and the gradual arrangement at less steep angles (Fig. 2). This tendency was observed especially in the long head.

Comparison between normal subjects ($n = 8$) and body-

builders ($n = 8$) revealed that fiber pennation angles (33 ± 16 vs. $15 \pm 6^\circ$ for long head and 19 ± 8 vs. $11 \pm 5^\circ$ for medial head) as well as muscle thickness (46 ± 18 vs. 26 ± 11 mm) were significantly greater ($P < 0.01$) in bodybuilders than in normal subjects. Similar results were obtained when muscle thickness was normalized by the upper arm length.

The relationship between muscle thickness and fiber pennation angles is shown in Fig. 3. Muscle thickness was normalized by upper arm lengths. Statistically significant positive correlations were observed with coefficients of $r = 0.884$ (long head) and $r = 0.833$ (medial head).

DISCUSSION

It is plausible that muscle thickness measurements can estimate muscle size and the degree of muscle hypertrophy because Martinson and Stokes (11) showed a significant correlation between linear measurements of ultrasound cross-sectional scans and muscle CSAs. Therefore, in the present study muscle thickness was used to represent muscle size.

Ultrasonic and manual measurements on cadavers showed that ultrasonography can well be used to measure muscle thickness and pennation angles with measurement errors < 1 mm and $< 1^\circ$, respectively. However, great care must be taken, especially in muscle thickness

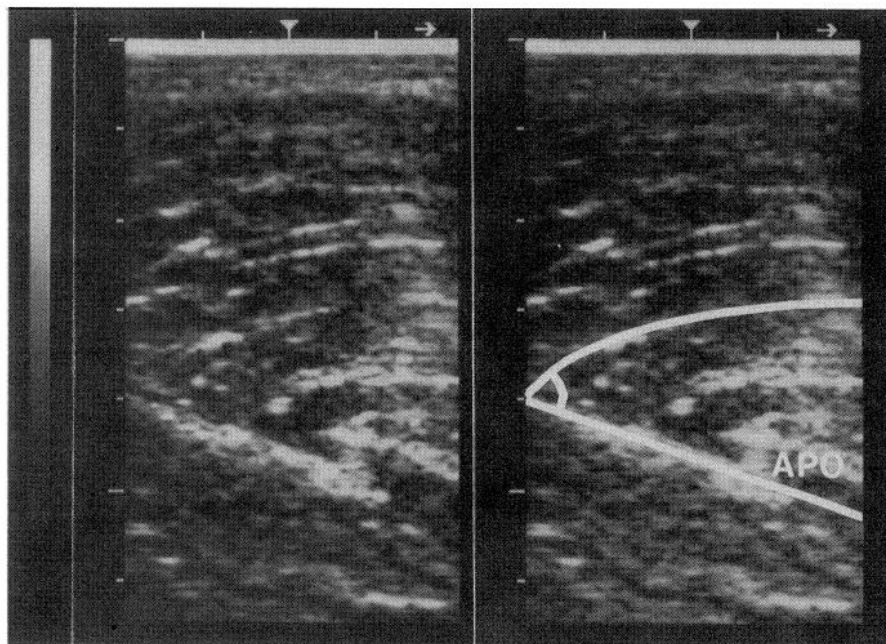


FIG. 2. Ultrasonographic image from 1 highly trained bodybuilder. Fascicles were arranged curvilinearly and fiber pennation angles were steeper where fascicles attached to APO rather than in midposition.

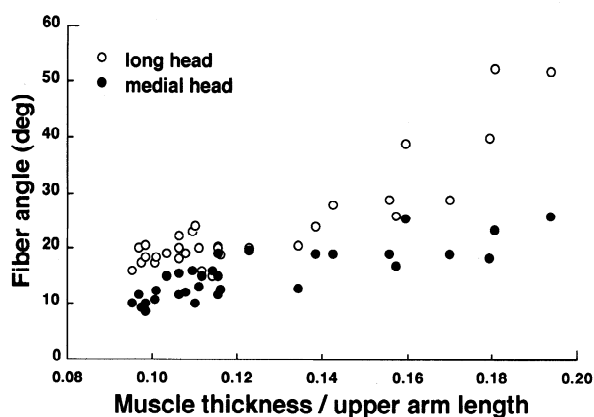


FIG. 3. Relationship between fiber pennation angles and ratio of muscle thickness to upper arm length. For both long and medial heads, statistically significant correlations were observed (long head, $r = 0.884$; medial head, $r = 0.833$).

measurement, because the thickness of a muscle varies greatly along its length, as shown in Fig. 1.

Fiber pennation angles measured in this study were similar to or greater than those in human cadavers in a previous study (1) and in this study. These discrepancies might be explained by the fact that in studies using cadaver muscles the samples were usually taken from older individuals of unknown health histories and after widely varying methods of fixation; thus shrinkage in muscle volume could have occurred (5).

The novel finding in this study was the interrelationship between muscle thickness and fiber pennation angles. This result would imply that muscle hypertrophy might accompany an increase in fiber angles, the possibility of which has been suggested in previous studies (9, 10, 14). Maxwell et al. (13) predicted that for a muscle of constant muscle length, fiber length, and fiber numbers, hypertrophy (i.e., increase in the diameter of fibers) must accompany increases in pennation angles and muscle thickness. The present results support their suggestion. However, Henriksson-Larsen et al. (6) used a methodology similar to that of the present study and found no correlation between muscle-fiber angulation and fiber CSA, and they concluded that fiber angulation was independent of fiber size, which was the opposite of the results of the present study. The discrepancy between the results might be attributed to the difference in subjects and muscles investigated. First, their subjects were females with average physical characteristics. A narrow range of variation in fiber CSAs might have resulted in the lack of statistical significance in the relationship between fiber size and fiber angulation. The present study used male subjects of various training statuses (from normal subjects to well-trained bodybuilders), which might have given significance to the relationship between muscle size and fiber angles. Second, Henriksson-Larsen et al. used vastus lateralis muscle in their study. Contribution of fiber angulation to muscle size might be different between vastus lateralis and triceps brachii. Further research will be required in this respect.

In normal subjects the fascicles (i.e., muscle fibers) were arranged linearly at similar angles, whereas in highly trained subjects with hypertrophied muscles the

fascicles were arranged curvilinearly, with fibers arising from the aponeurosis at steeper angles. Considering that greater pennation angles would result in more contractile material attached to a larger area of the tendon (9), the curvilinear arrangement in the muscle fibers might be a result of muscle hypertrophy. This arrangement would not significantly increase anatomical muscle CSA, which would make the relationship between muscle CSA and muscle force different from the relationship in muscles with linear pennation. This hypothesis might be able to explain the results of Jones and Rutherford (9), who found that resistance training resulted in an increase in force but not so much in muscle CSA, and of Maughan et al. (12), who found that an inverse relationship existed between CSA and strength per unit CSA.

Physiological CSA of muscle is calculated by (16)

$$\text{physiological CSA} = (\text{MV/FL}) \cos \alpha$$

where MV is muscle volume and FL is muscle-fiber length. Cosine of pennation angle α normalizes fiber angulation to the line of action of the muscle. If muscle hypertrophy accompanied the increase in fiber pennation angles, then this change would be disadvantageous for the force development. However, an increase in pennation angles would cause more contractile material to be attached to the tendon, as stated above, which would increase muscle volume and consequently increase physiological CSA. These hypotheses need to be tested by longitudinal training studies.

It must be noted that the pennation angles measured in the present study were those between the aponeurosis and the fascicles. The pennation angles may not be accurate vectors for predicting the effect of angulation of fibers on force transfer to the tendon if the aponeurosis lies at an angle to the line of action of the muscle. But, according to visual observation on human cadavers, the angle of the aponeurosis on the deep border to the long axis of the long head was small compared with that of the opposite aponeurosis; thus the error in predicting the impact of fiber angulation on the force transfer from the pennation angle would be small, at least for the long head. However, information on the angle between the aponeurosis and the line of action of the muscle must be obtained to investigate the force transfer accurately.

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